

```
import os
print(os.listdir())
```

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```
DATA_PATH = '/content/drive/MyDrive/Olist dataset/'

import pandas as pd

orders = pd.read_csv(DATA_PATH + 'olist_orders_dataset.csv')
reviews = pd.read_csv(DATA_PATH + 'olist_order_reviews_dataset.csv')
customers = pd.read_csv(DATA_PATH + 'olist_customers_dataset.csv')
items = pd.read_csv(DATA_PATH + 'olist_order_items_dataset.csv')
products = pd.read_csv(DATA_PATH + 'olist_products_dataset.csv')
translation = pd.read_csv(DATA_PATH + 'product_category_name_translation.csv')

print("All loaded successfully")
```

All loaded successfully

```
for name, df in [('orders', orders), ('reviews', reviews), ('customers', customers),
                 ('items', items), ('products', products)]:
    print(f"--- {name} ---")
    print("Shape:", df.shape)
    if 'order_id' in df.columns:
        print(" Unique order_id count:", df['order_id'].nunique(), "vs total rows:", len(df))
    print()
```

```
--- orders ---
Shape: (99441, 8)
Unique order_id count: 99441 vs total rows: 99441

--- reviews ---
Shape: (99224, 7)
Unique order_id count: 98673 vs total rows: 99224

--- customers ---
Shape: (99441, 5)

--- items ---
Shape: (112650, 7)
Unique order_id count: 98666 vs total rows: 112650

--- products ---
Shape: (32951, 9)
```

```
reviews_clean = reviews.sort_values('review_creation_date').drop_duplicates(
    subset='order_id', keep='last'
)
print("Reviews before:", len(reviews), "after dedupe:", len(reviews_clean))
```

Reviews before: 99224 after dedupe: 98673

```
items_products = items.merge(products, on='product_id', how='left')
items_products = items_products.merge(translation, on='product_category_name', how='left')

order_category = items_products.groupby('order_id').agg(
    product_category_name_english=('product_category_name_english', 'first'),
    item_count=('order_item_id', 'count'),
    total_price=('price', 'sum')
).reset_index()

print("order_category rows:", len(order_category))
```

order_category rows: 98666

```
master = orders.merge(customers, on='customer_id', how='left')
master = master.merge(reviews_clean[['order_id', 'review_score']], on='order_id', how='left')
master = master.merge(order_category, on='order_id', how='left')
```

```
print("Orders count:", len(orders))
print("Master count:", len(master))
```

```
Orders count: 99441
Master count: 99441
```

```
master['order_estimated_delivery_date'] = pd.to_datetime(master['order_estimated_delivery_date'])
master['order_delivered_customer_date'] = pd.to_datetime(master['order_delivered_customer_date'])

master['delivery_flag'] = 'Delivered'
master.loc[master['order_status'].isin(['canceled', 'unavailable']), 'delivery_flag'] = 'Not Delivered'
master.loc[master['order_delivered_customer_date'].isna() & (master['delivery_flag'] == 'Delivered'), 'delivery_flag'] = 'Missi

print(master['delivery_flag'].value_counts())
```

```
delivery_flag
Delivered      96470
Missing Date    1737
Not Delivered   1234
Name: count, dtype: int64
```

```
delivered = master[master['delivery_flag'] == 'Delivered'].copy()

delivered['Days_Difference'] = (
    delivered['order_estimated_delivery_date'] - delivered['order_delivered_customer_date']
).dt.days

def classify(days):
    if days >= 0:
        return 'On Time'
    elif days >= -5:
        return 'Late'
    else:
        return 'Super Late'

delivered['delivery_status'] = delivered['Days_Difference'].apply(classify)

print(delivered['delivery_status'].value_counts())
print(delivered['delivery_status'].value_counts(normalize=True) * 100)
```

```
delivery_status
On Time      88644
Super Late   4211
Late         3615
Name: count, dtype: int64
delivery_status
On Time      91.887633
Super Late    4.365088
Late         3.747279
Name: proportion, dtype: float64
```

```
state_late_pct = (
    delivered.groupby('customer_state')['delivery_status']
    .apply(lambda x: (x != 'On Time').mean() * 100)
    .reset_index(name='pct_late')
    .sort_values('pct_late', ascending=False)
)
print(state_late_pct)
```

```
customer_state  pct_late
1              AL  23.929471
9              MA  19.665272
16             PI  15.966387
5              CE  15.324472
24             SE  15.223881
4              BA  14.035627
18             RJ  13.473684
26             TO  12.773723
13             PA  12.367865
7              ES  12.230576
21             RR  12.195122
11             MS  11.554922
14             PB  11.025145
15             PE  10.797238
19             RN  10.759494
23             SC  9.757473
8              GO  8.175779
22             RS  7.148204
6              DF  7.067308
```

12	MT	6.772009
25	SP	5.894700
10	MG	5.610358
17	PR	4.996953
3	AP	4.477612
2	AM	4.137931
0	AC	3.750000
20	RO	2.880658

```
state_volume = delivered.groupby('customer_state').size().reset_index(name='order_count')
state_summary = state_late_pct.merge(state_volume, on='customer_state')
print(state_summary.sort_values('pct_late', ascending=False))
```

	customer_state	pct_late	order_count
0	AL	23.929471	397
1	MA	19.665272	717
2	PI	15.966387	476
3	CE	15.324472	1279
4	SE	15.223881	335
5	BA	14.035627	3256
6	RJ	13.473684	12350
7	TO	12.773723	274
8	PA	12.367865	946
9	ES	12.230576	1995
10	RR	12.195122	41
11	MS	11.554922	701
12	PB	11.025145	517
13	PE	10.797238	1593
14	RN	10.759494	474
15	SC	9.757473	3546
16	GO	8.175779	1957
17	RS	7.148204	5344
18	DF	7.067308	2080
19	MT	6.772009	886
20	SP	5.894700	40494
21	MG	5.610358	11354
22	PR	4.996953	4923
23	AP	4.477612	67
24	AM	4.137931	145
25	AC	3.750000	80
26	RO	2.880658	243

```
import plotly.express as px

fig = px.bar(
    state_summary.sort_values('pct_late', ascending=False),
    x='customer_state', y='pct_late',
    hover_data=['order_count'],
    title='% Late Deliveries by State',
    labels={'pct_late': '% Late', 'customer_state': 'State'})
fig.show()
```

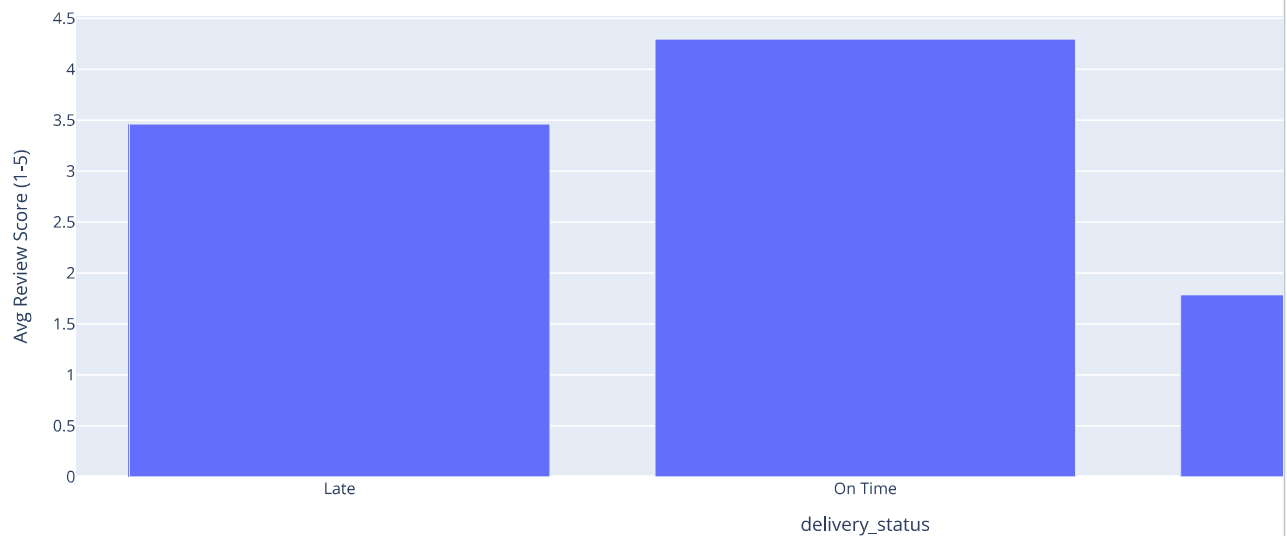
[Show hidden output](#)

```
avg_score_by_status = delivered.groupby('delivery_status')['review_score'].mean().reset_index()
print(avg_score_by_status)
```

	delivery_status	review_score
0	Late	3.460482
1	On Time	4.294001
2	Super Late	1.784999

```
fig2 = px.bar(avg_score_by_status, x='delivery_status', y='review_score',
    title='Average Review Score by Delivery Status',
    labels={'review_score': 'Avg Review Score (1-5)'})
fig2.show()
```

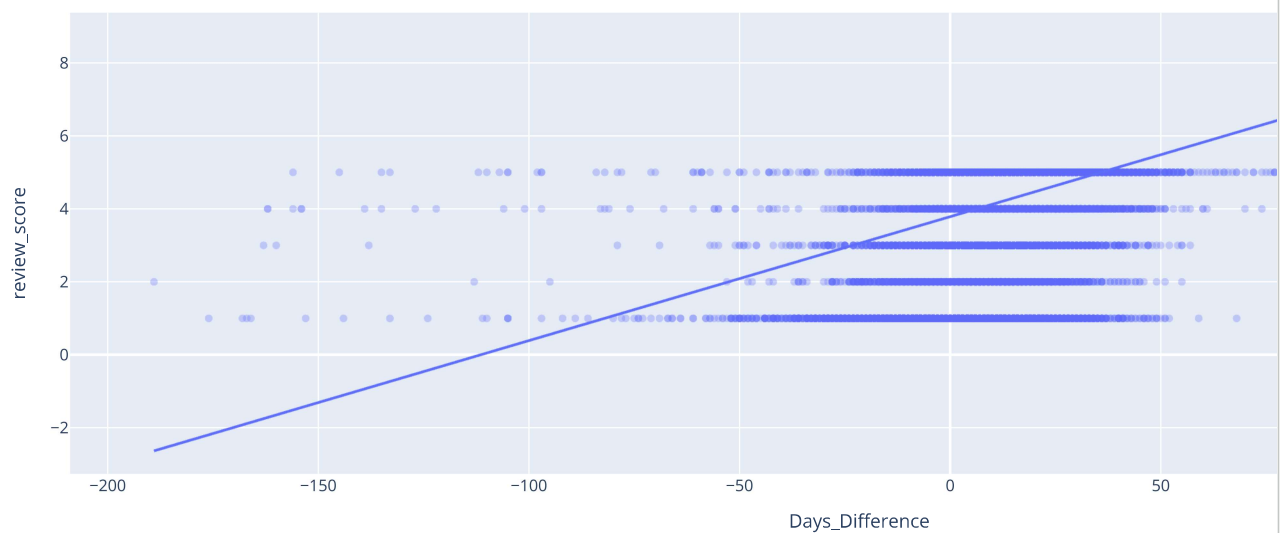
Average Review Score by Delivery Status



```
!pip install statsmodels -q

fig3 = px.scatter(delivered, x='Days_Difference', y='review_score',
                  trendline='ols', opacity=0.3,
                  title='Delivery Delay (Days) vs Review Score')
fig3.show()
```

Delivery Delay (Days) vs Review Score



```
print(delivered['product_category_name_english'].value_counts().head(10))
```

```
product_category_name_english
bed_bath_table      9171
health_beauty       8608
sports_leisure       7492
computers_accessories 6501
furniture_decor      6213
housewares           5695
watches_gifts        5470
telephony            4076
auto                 3793
```

```
toys          3779
Name: count, dtype: int64
```

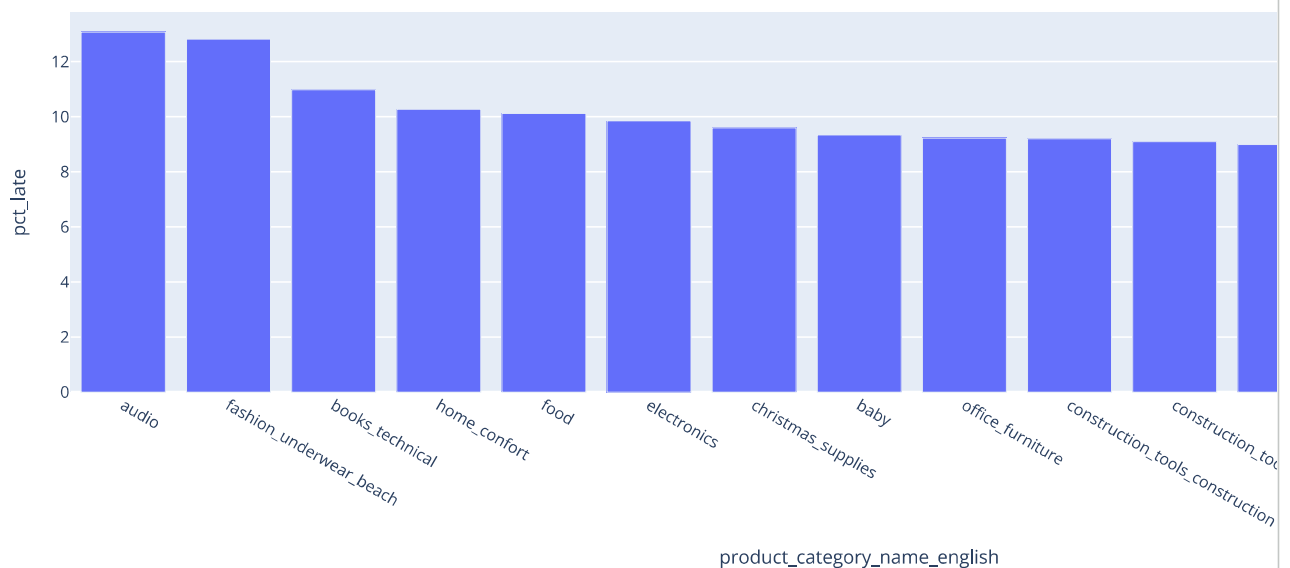
```
category_delay = (
    delivered.groupby('product_category_name_english')['delivery_status']
    .apply(lambda x: (x != 'On Time').mean() * 100)
    .reset_index(name='pct_late')
)
category_volume = delivered.groupby('product_category_name_english').size().reset_index(name='order_count')
category_summary = category_delay.merge(category_volume, on='product_category_name_english')

# Filter out tiny categories for reliability, then sort
category_summary = category_summary[category_summary['order_count'] >= 50].sort_values('pct_late', ascending=False)
print(category_summary.head(15))
```

	product_category_name_english	pct_late	order_count
4	audio	13.081395	344
33	fashion_underwear_beach	12.820513	117
10	books_technical	10.980392	255
47	home_comfort	10.270270	370
36	food	10.114943	435
26	electronics	9.852413	2507
12	christmas_supplies	9.600000	125
6	baby	9.337676	2763
57	office_furniture	9.229535	1246
17	construction_tools_construction	9.203297	728
18	construction_tools_lights	9.090909	231
43	health_beauty	8.991636	8608
56	musical_instruments	8.881579	608
7	bed_bath_table	8.843092	9171
5	auto	8.647509	3793

```
fig4 = px.bar(category_summary.head(15), x='product_category_name_english', y='pct_late',
               hover_data=['order_count'],
               title='Top 15 Product Categories by % Late Delivery (min 50 orders)')
fig4.show()
```

Top 15 Product Categories by % Late Delivery (min 50 orders)



```
delivered.to_csv('cleaned_delivery_data.csv', index=False)

from google.colab import files
files.download('cleaned_delivery_data.csv')
```

